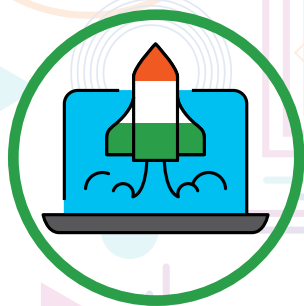




18 INDIAN LOCAL PROBLEMS THAT ARE BEING SOLVED BY TECH

India has always had unique problems as well as solutions to those problems. Globally, with everyone looking to technology as an easy fix to problems that have plagued us for years, we look at some uniquely Indian solutions.

Indian tech startups and companies have begun creating and molding technology to cater to local issues that are uniquely Indian and are solving them at the grassroots level.

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01

FARMER SUICIDES

About 70 per cent of the rural population of India relies on agriculture as their source of income. So how can technology help? A product called MATT by Nebulaa Innovations attempts to solve the problem of wholesalers falsely claiming low quality of produce in order to short change farmers. Using image processing and AI, MATT just needs someone to click a picture of the produce and that gives it a rating and a valuation. Because over 75 per cent of farmers in India own one hectare of land or less, it isn't viable for them to invest in machinery such as tractors. farMart is an on-demand agri machinery renting platform for farmers, by farmers. Not only do farmers with small plots get the help of machinery, those who own machines earn extra income, in a win-win outcome.

02

HOMELESSNESS

Although homelessness is not a uniquely Indian problem, we do have a huge problem of people living below the poverty line – about 29.6% of population in fact. Although most of us have been conditioned to ignore the reality, there

are homeless people on the roads, pavements, hume-pipes, under staircases and flyovers, at railway stations, places of worship... everywhere! A startup called Billionbricks is attempting to help the homeless by mass manufacturing good-quality tents. Called WeatherHYDE, the tent can accommodate a family of two adults and three children.

03

MANUAL SCAVENGING

Since 2017, one person dies every five days (on average) while cleaning sewers or septic tanks across the country. A Thiruvananthapuram-based startup called Genrobotics, uses a spider shaped robot called Bandicoot, to go down into manholes to clean sewers. The 50-kg, pneumatic-powered robot is controlled via remote control, and cleans using a 360-degree sweep motion. The Bandicoot additionally utilises machine learning (ML) and artificial intelligence (AI) to figure out the amount of unclogging that is required for each sewer.

04

POWER SHORTAGE

In a diverse country like India with varying weather

across different states, power usage varies wildly. This disparity of consumption leads to disparity in supply, despite the country having enough power for everyone's needs. IEX (Indian Energy Exchange) attempts to solve this by providing an efficient electronic power trading platform. This enables those states with surplus to sell to those with a deficit. Currently, IEX now has a pan-India presence with over 4000 participants that include power generators, industrial consumers and companies that distribute power. Another startup called Aatapaha helps save power by using smart street lighting that automatically dims streetlights when there are no vehicles on the road.

05

LANGUAGE BARRIER

A very unique problem we Indians have is being able to understand one another, because of how many languages we speak. This language barrier is cited as one of the reasons for low penetration of smartphones in India. Indus OS is the world's first regional smartphone OS which allows Indians to interact and communicate in their own native language. The